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TELESCOPING RODS FOR USE WITH VANITY MIRRORS

Abstract

Telescoping rods including: a core tube having a first end and a second end; a first pulley mounted inside the core tube at the first end; a second pulley mounted inside the core tube at the second end; a belt mounted around and between the first pulley and the second pulley; a first arm coupled to the belt, partially located inside the core tube, and extending outward of the first end of the core tube; and a second arm coupled to the belt, partially located inside the core tube, and extending outward of the second end of the core tube. In some implementations, the belt contains teeth in an inside surface of the belt, the first pulley contains teeth that engage the teeth of the belt, and the second pulley contains teeth that engage the belt.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application: is a continuation of International Patent Application No. PCT/US2025/013266, filed Jan. 27, 2025, which claims the benefit of U.S. Provisional Patent Application No. 63/625,614, filed Jan. 26, 2024; and is a continuation-in-part of U.S. patent application Ser. No. 18/630,565, filed Apr. 9, 2024, which is a divisional of U.S. patent application Ser. No. 18/072,052, filed Nov. 30, 2022, now U.S. Pat. No. 11,980,303, which is a continuation of U.S. patent application Ser. No. 17/537,303, filed Nov. 29, 2021, now U.S. Pat. No. 11,576,507, which is a continuation-in-part of U.S. patent application Ser. No. 16/548,725, filed Aug. 22, 2019, now U.S. Pat. No. 11,209,609, and which claims the benefit of U.S. Provisional Application No. 62/722,749, filed on Aug. 24, 2018, each of which is hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD
[0002] The present disclosure relates generally to vanity mirrors.
BACKGROUND
[0003] Vanity mirrors are reflective devices that are typically used for personal grooming, shaving, applying makeup, or the like.
[0004] Vanity mirrors are available in various configurations, including wall-mounted, free-standing, and hand-held. A common problem, however, with each of these configurations is that users of the mirror must reposition themselves to obtain the desired distance from the reflective surface and the requisite height for the reflective surface to be viewable. This repositioning is particularly undesirable when the user is required to bend over to reach the desired height and distance, which may cause back and neck pain. In the case of a hand-held mirror, the user may be able to reposition the mirror but must hold the mirror at the desired location, thereby reducing their ability to perform the required grooming tasks.
[0005] Accordingly, new adjustable vanity mirrors are desirable.
SUMMARY
[0006] In accordance with some embodiments, telescoping rods for use with vanity mirrors are provided.
[0007] In some embodiments, telescoping rods are provided, the telescoping rods comprising: a core tube having a first end and a second end; a first pulley mounted inside the core tube at the first end; a second pulley mounted inside the core tube at the second end; a belt mounted around and between the first pulley and the second pulley; a first arm coupled to the belt, partially located inside the core tube, and extending outward of the first end of the core tube; and a second arm coupled to the belt, partially located inside the core tube, and extending outward of the second end

of the core tube.
[0008] In some of these embodiments, the belt contains teeth in an inside surface of the belt, the first pulley contains teeth that engage the teeth of the belt, and the second pulley contains teeth that engage the belt.
[0009] In some of these embodiments, the telescoping rods further comprise a first clamp that couples the belt to the first arm and a second clamp that couples the belt to the second arm.
[0010] In some of these embodiments, the telescoping rods further comprise a middle tube that contains the core tube, wherein the middle tube has a first end and a second end.
[0011] In some of these embodiments, the telescoping rods further comprise a cable positioned between to core tube and the middle tube.
[0012] In some of these embodiments, the telescoping rods further comprise a top tube that at least partially fits inside the middle tube, extends outward from the middle tube at the first end of the middle tube, and is coupled to the first arm.
[0013] In some of these embodiments, the telescoping rods further comprise first bushing located on an outside surface of the top tube at the second end of the top tube.
[0014] In some of these embodiments, the first bushing comprises two parts.
[0015] In some of these embodiments, a middle portion of the first bushing has a larger diameter than another portion of the first bushing.
[0016] In some of these embodiments, the telescoping rods further comprise a second bushing located at the first end of the middle tube between the top tube and the middle tube.
[0017] In some of these embodiments, the second bushing includes at least one raised surface that presses on an inside surface of the middle tube and cause a portion of the second bushing to bend and apply pressure to an outside surface of the top tube.
[0018] In some of these embodiments, the telescoping rods further comprise a top cap that couples the first arm and the top tube to a mirror.
[0019] In some of these embodiments, the telescoping rods further comprise a bottom tube that at least partially extends over the middle tube, extends away from the middle tube at the second end of the middle tube, and is coupled to the second arm.
[0020] In some of these embodiments, the telescoping rods further comprise a third bushing located at a first end of the bottom tube between the bottom tube and the middle tube.
[0021] In some of these embodiments, the third bushing includes at least one raised surface that presses on an inside surface of the bottom tube and cause a portion of the third bushing to bend and apply pressure to an outside surface of the middle tube.
[0022] In some of these embodiments, the telescoping rods further comprise a pad coupled to the middle tube that presses on the bottom tube to slow movement between the middle tube and the bottom tube.
[0023] In some of these embodiments, the pad is longer in a first dimension along the axis of the middle tube than it is wide in a second dimension that is perpendicular to the first dimension.
[0024] In some of these embodiments, the telescoping rods further comprise a bottom cap that couples the second arm and the bottom tube to a base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0025] FIG. **1** illustrates a front view of an example of a mirror assembly with the telescoping rod extended and trigger arm engaged in accordance with some embodiments.
[0026] FIG. **2** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod extended, trigger arm engaged, and hanger bracket retracted in accordance with some embodiments.

[0027] FIG. **3** illustrates a rear view of an example of a mirror assembly with the telescoping rod extended and hanger bracket retracted in accordance with some embodiments.
[0028] FIG. **4** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod extended, trigger arm engaged, and hanger bracket retracted in accordance with some embodiments.
[0029] FIG. **5** illustrates a side view of an example of a mirror assembly with the telescoping rod extended, trigger arm engaged, and hanger bracket retracted in accordance with some embodiments.
[0030] FIG. **6** illustrates a front view of an example of a mirror assembly with the telescoping rod retracted and trigger arm disengaged in accordance with some embodiments.
[0031] FIG. **7** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, and hanger bracket retracted in accordance with some embodiments.
[0032] FIG. **8** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, and hanger bracket retracted in accordance with some embodiments.
[0033] FIG. **9** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket extended, and reflective surfaces partially rotated in accordance with some embodiments.
[0034] FIG. **10** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket extended, and reflective surfaces partially rotated in accordance with some embodiments.
[0035] FIG. **11**A illustrates a front view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0036] FIG. **11**B illustrates a front view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0037] FIG. **12**A illustrates a bottom perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0038] FIG. **12**B illustrates a bottom perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0039] FIG. **13**A illustrates a front perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0040] FIG. **13**B illustrates a front perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0041] FIG. **14** illustrates a front view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, and reflective surfaces partially rotated in accordance with some embodiments.
[0042] FIG. **15** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.
[0043] FIG. **16** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.
[0044] FIG. **17** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.
[0045] FIG. **18** illustrates a side view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.
[0046] FIG. **19** illustrates a front view of an example of a mirror assembly with the telescoping rod

extended and trigger arm engaged in accordance with some embodiments.
[0047] FIG. **20** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod extended, trigger arm engaged, and hanger bracket retracted in accordance with some embodiments.
[0048] FIG. **21** illustrates a rear view of an example of a mirror assembly with the telescoping rod extended and hanger bracket retracted in accordance with some embodiments.
[0049] FIG. **22** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod extended, trigger arm engaged, and hanger bracket retracted in accordance with some embodiments.
[0050] FIG. **23** illustrates a side view of an example of a mirror assembly with the telescoping rod extended, trigger arm engaged, and hanger bracket retracted in accordance with some embodiments.
[0051] FIG. **24** illustrates a front view of an example of a mirror assembly with the telescoping rod retracted and trigger arm disengaged in accordance with some embodiments.
[0052] FIG. **25** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, and hanger bracket retracted in accordance with some embodiments.
[0053] FIG. **26** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, and hanger bracket retracted in accordance with some embodiments.
[0054] FIG. **27** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket extended, and reflective surfaces partially rotated in accordance with some embodiments.
[0055] FIG. **28** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket extended, and reflective surfaces partially rotated in accordance with some embodiments.
[0056] FIG. **29**A illustrates a front view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0057] FIG. **29**B illustrates a front view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0058] FIG. **30**A illustrates a bottom perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0059] FIG. **30**B illustrates a bottom perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0060] FIG. **31**A illustrates a front perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0061] FIG. **31**B illustrates a front perspective view of an example of a mirror assembly with the hanger bracket extended and engaged over a door or ledge in accordance with some embodiments.
[0062] FIG. **32** illustrates a front view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, and reflective surfaces partially rotated in accordance with some embodiments.
[0063] FIG. **33** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.
[0064] FIG. **34** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.
[0065] FIG. **35** illustrates a rear perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.

[0066] FIG. **36** illustrates a side view of an example of a mirror assembly with the telescoping rod retracted, trigger arm disengaged, hanger bracket retracted, and reflective surfaces partially rotated in accordance with some embodiments.
[0067] FIG. **37** illustrates a front view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket retracted, and housing rotated in accordance with some embodiments.
[0068] FIG. **38** illustrates a side view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket retracted, and housing rotated in accordance with some embodiments.
[0069] FIG. **39** illustrates a front perspective view of an example of a mirror assembly with the telescoping rod retracted, trigger arm engaged, hanger bracket retracted, and housing rotated in accordance with some embodiments.
[0070] FIG. **40** illustrates an example of a core telescope assembly, including an upper arm, two pulleys, a belt, two clamps, and a lower arm, of a telescoping rod in accordance with some embodiments.
[0071] FIG. **41** illustrates an example of an upper arm of a telescoping rod in accordance with some embodiments.
[0072] FIG. **42** illustrates an example of a belt of a telescoping rod in accordance with some embodiments.
[0073] FIG. **43** illustrates an example of a clamp of a telescoping rod in accordance with some embodiments.
[0074] FIG. **44** illustrates an example of a pulley of a telescoping rod in accordance with some embodiments.
[0075] FIG. **45** illustrates an example of a lower arm of a telescoping rod in accordance with some embodiments.
[0076] FIG. **46** illustrates an example of core tube over the components of FIG. **40** in accordance with some embodiments.
[0077] FIG. **47** illustrates an example of a core tube in accordance with some embodiments.
[0078] FIG. **48** illustrates an example of the components of FIG. **46** with the addition of a first end cap in accordance with some embodiments.
[0079] FIG. **49** illustrates an example of the first end cap of FIG. **48** in accordance with some embodiments.
[0080] FIG. **50** illustrates an example of the components of FIG. **48** with the addition of a cable in accordance with some embodiments.
[0081] FIG. **51** illustrates an example of the components of FIG. **50** inside a middle tube in accordance with some embodiments.
[0082] FIG. **52** illustrates an example of the middle tube of FIG. **51** in accordance with some embodiments.
[0083] FIG. **53** illustrates an example of the components of FIG. **51** with a top tube in accordance with some embodiments.
[0084] FIG. **54** illustrates an example of the top tube in accordance with some embodiments.
[0085] FIG. **55** illustrates an example of the top tube with a first bushing in accordance with some embodiments.
[0086] FIG. **56** illustrates an example of the first bushing in accordance with some embodiments.
[0087] FIG. **57** illustrates an example of the components of FIG. **55** along with a second bushing in accordance with some embodiments.
[0088] FIG. **58** illustrates an example of the second bushing in accordance with some embodiments.
[0089] FIG. **59** illustrates an example of the components of FIG. **57** along with a third bushing in accordance with some embodiments.

[0090] FIG. **60** illustrates an example of the third bushing in accordance with some embodiments.
[0091] FIG. **61** illustrates an example of a friction pad in accordance with some embodiments.
[0092] FIG. **62** illustrates an example of the friction pad in accordance with some embodiments.
[0093] FIG. **63** illustrates an example of a top tube, a middle tube, and a bottom tube of a telescoping rod in accordance with some embodiments.
[0094] FIG. **64** illustrates an example of the bottom tube in accordance with some embodiments.
[0095] FIG. **65** illustrates an example of the top tube, the middle tube, and the bottom tube along with a second end cap in accordance with some embodiments.
[0096] FIG. **66** illustrates an example of the second end cap in accordance with some embodiments.
[0097] FIG. **67** illustrates an example of the top tube, the middle tube, and the bottom tube along with a third end cap in accordance with some embodiments.
[0098] FIG. **68** illustrates an example of the third end cap in accordance with some embodiments.
[0099] FIG. **69** illustrates an example of a suction cup having a suction release mechanism in accordance with some embodiments.
[0100] FIG. **70** illustrates an example of a pulley of a telescoping rod have a groove for an o-ring along with an o-ring and bumps on an adjacent rod in accordance with some embodiments.
[0101] FIG. **71** illustrates an example of a bushing as shown in FIG. **60** inside a tube in accordance with some embodiments.
[0102] FIG. **72** illustrates an example of a modified bushing of FIG. **56** in accordance with some embodiments.

DETAILED DESCRIPTION

[0103] In accordance with some embodiments, adjustable vanity mirrors are provided.
[0104] In some embodiments, a mirror assembly comprising a mirror, suction cup base with a hanger bracket, swivel joint, adjustable telescoping rod, and light source with diffuser screen is provided.
[0105] In some embodiments, a hands-free, adjustable telescoping magnifying mirror with a swivel joint, adjustable telescoping rod, and light source with diffuser screen is provided. The mirror can be secured to a horizontal or vertical surface using a suction cup base or can be hung over a door or ledge using a retractable hanger bracket, in some embodiments. In some embodiments, a telescoping rod is attached to the suction cup base. The mirror can be extended from the base by extending the telescoping rod and the mirror assembly can stand freely without falling over, whether the suction cup base is engaged or not, in some embodiments. In some embodiments, the telescoping rod can have a base stage and four additional stages, wherein the base stage has a certain diameter, stages one and two have progressively decreasing diameters, and stages three and four have progressively increasing diameters such that the diameter of stage four can be equal to the diameter of the base stage. In some embodiments, this multi-stage configuration permits the use of a larger connection point for attachment of a joint, such as a swivel joint, to secure the telescoping rod to the mirror housing, thereby increasing the stability and strength of the joint while also allowing stage four of the telescoping rod to cover the joint, thereby additionally providing aesthetic benefits. A smooth disc can also be supplied to increase stability of the mirror assembly by removably mounting the suction cup base to the disc, giving the base a larger footprint, in some embodiments.
[0106] In some embodiments, the swivel joint provides several advantages, including increased repositionability and increased portability and convenience. Additionally, in some embodiments, the light source with diffuser screen provides several advantages, including providing a uniform light glow and automatic deactivation when the mirror assembly is using battery power.
[0107] In some embodiments, the mirror assembly has a first reflective surface disposed on a first side of the mirror and second reflective surface disposed on a second side of the mirror. The first reflective surface and second reflective surface can be magnifying or non-magnifying and can have

different levels of magnification than one another, in some embodiments. The mirror can also have a lever that when triggered rotates the first reflective surface and second reflective surface relative to light source or sources about an axis normal to the vertical axis of the telescoping rod, for example, thereby allowing a user to view the reflective surface that provides the desired magnification level, in some embodiments. In some embodiments, the mirror can include a clamp brake that adds friction to the rotation of the first and second reflective surfaces, thereby enhancing the user's ability to make back-and-forth adjustments to the position of the reflective surfaces and increasing the ability of the first and second reflective surfaces to remain in the desired position.

[0108] In some embodiments, mirrors as described herein solve one or more problems of prior mirrors by providing a hands-free, adjustable mirror assembly with a suction cup base, hanger bracket, swivel joint, telescoping rod, and light source with diffuser screen. In some embodiments, the suction cup base allows the user to position the mirror assembly at the desired distance, height, and angle from the user while maintaining the stability of the mirror assembly. In some embodiments, the swivel joint allows the user to rotate the mirror about the axis of the swivel joint, thereby allowing a user of the mirror to achieve the desired mirror position in relation to the position of the user. For example, in some embodiments, if a user is seated while using the mirror, the swivel joint allows the user to rotate the mirror to a height that permits the user to more comfortably view the mirror at the angle desired by the user. The swivel joint can be operated by, for example, rotating the mirror about the axis of the swivel joint, in some embodiments. The swivel joint also permits a user to adjust the position of the mirror in a one-step process that does not require the user to fold the mirror such that the light source no longer faces the user. In some embodiments, the swivel joint can include a spring-loaded pin mechanism that permits the user to lock the mirror in the desired position. In some embodiments, the swivel joint also permits a user to rotate the mirror housing for more compact storage and transport of the mirror assembly, thereby increasing portability and convenience, in some embodiments. In some embodiments, the light source can include a diffuser screen to diffuse light emitted by the light source thereby providing a uniform light glow. In some embodiments, the light source can automatically deactivate after a certain period of time, for example, 20 minutes, thereby conserving battery power when the mirror assembly is not plugged in to an electrical outlet. In some embodiments, when the light source is set to the user's desired dimmer setting and the light source is deactivated and later reactivated, the light source will return to the previous dimmer setting. In some embodiments, the hanger bracket permits the user to hang the mirror assembly at the desired height and position over a door or ledge while maintaining the stability of the mirror assembly. These features of some embodiments permit a user of the corresponding embodiments to stand or sit, as desired, and permit the user to move the mirror assembly to the desired location rather than bending over to reach the desired distance from the reflective surface, thereby potentially avoiding causing back and neck pain. The ability to hang the present mirror assembly over a door or ledge also provides the user with the ability to utilize the mirror assembly in a variety of locations, in some embodiments.

[0109] FIGS. **1-18** illustrate mirror assemblies in accordance with some embodiments. In some embodiments, a mirror assembly **10** comprises a mirror **20**, a suction cup base assembly **30**, and a telescoping rod **40**. Mirror **20** can have a first reflective surface **21** on a first side and a second reflective surface **22** on a second side, in some embodiments. In some embodiments, there can be one or more light sources **23** disposed at a periphery of first reflective surface **21** or second reflective surface **22**. The light source or sources **23** can be powered by a battery (e.g., a rechargeable battery) or can be plugged into an electrical outlet, for example. The light source or sources **23** can be activated and deactivated by a switch **24**, or the like, which can be located adjacent the light source **23** on the front side of housing **27**, on the reverse side of housing **27**, or in the suction cup base assembly **30**, for example. Switch **24** can be a push-button toggle switch, for example, or can be a rotatable knob that allows a user to adjust the intensity of light emitted by light source **23** by rotating the knob in a clockwise or counterclockwise direction. Housing **27** can have a socket for use in a ball joint by which a first end **41** of telescoping rod **40** can be secured to housing **27**. The mirror **20** can also have a lever **26** that when triggered rotates reflective surfaces **21** and **22** relative to light source or sources **23** about an axis normal to the vertical axis of telescoping rod **40**, for example, thereby allowing a user to view the reflective surface that provides the desired magnification level. In some embodiments, other methods and axes of rotation can be used to rotate reflective surfaces **21** and **22**. Each reflective surface **21** and **22** can be non-magnifying or can be magnifying, and each can provide level a of magnification of, for example, 1×, 3×, 5×, 7×, 10×, 15×, or 20×.

[0110] In some embodiments, suction cup base assembly **30** is substantially hemispherical in shape, comprising a round, flexible, concave diaphragm **34** (i.e., a suction cup) on the front side of the planar surface of the hemisphere, as illustrated in FIGS. **13**A, **13**B, **31**A, and **31**B. In some embodiments, the suction cup base assembly **30** can comprise any suitable number of suction cups. In some embodiments, the suction cup base assembly **30** comprises only one suction cup **34** in order to reduce the size and weight of the mirror assembly **10**, thereby increasing portability. The suction cup **34** can be used to removably mount the mirror assembly **10** to a horizontal or vertical surface, in some embodiments. In some embodiments, the suction cup **34** can be a locking suction cup or a non-locking suction cup. In some embodiments, a housing portion **31** is disposed on the reverse side of the planar surface of the hemisphere. The housing portion **31** can be made of any suitable material, such as a water-, abrasion-, and stain-resistant material, such as plastic, in some embodiments. In some embodiments, a material such as plastic will resist discoloration and rust that can occur due to conditions in which the mirror assembly can be utilized, e.g., in a damp environment such as a bathroom, adjacent to a water source such as a sink, or near substances that can stain or discolor such as makeup or other grooming products. In some embodiments, the plastic material from which the housing portion **31** of the suction cup base assembly **30** is preferably comprised can also be flexible such that it will not crack or break if the mirror assembly **10** is dropped. If the suction cup **34** is a locking suction cup, the housing portion **31** of the suction cup base assembly **30** can also include a trigger arm **33** for engaging the suction cup **34** such that it becomes removably mounted to a horizontal or vertical surface, in some embodiments. The trigger arm **33** can be comprised of any suitable material, in some embodiments. In some embodiments, due to the conditions in which the mirror assembly **10** can be utilized, e.g., in a damp environment such as a bathroom, or adjacent to a water source such as a sink, the trigger arm **33** can be selected to be comprised of a water- and rust-resistant material such as stainless steel or aluminum.

[0111] In some embodiments, the suction cup base assembly **30** can also include a hanger bracket **32**. The hanger bracket **32** can provide support for the mirror assembly **10** to be hung in an inverted manner on a door, a ledge, or the like, in some embodiments. To permit the mirror assembly **10** to be hung in an inverted manner from a door, ledge, or the like, the hanger bracket **32** can have an internal dimension sufficient to accommodate the minimum and maximum standard thickness of an interior door, in some embodiments. The hanger bracket **32** can be shaped as shown, for example, in FIGS. **9**, **10**, **13**A, **13**B, or can be shaped in any manner suitable to allow it to support the mirror assembly **10** over a door, ledge, or the like, in some embodiments. The hanger bracket **32** can be made of any suitable material, such metal (e.g., stainless steel) so that it can adequately support the mirror assembly **10** and is thin enough to permit the door on which the mirror assembly **10** can be hung to be closed, in some embodiments. In some embodiments, due to the conditions in which the mirror assembly **10** can be utilized, e.g., in a damp environment such as a bathroom, or adjacent to a water source such as a sink, the hanger bracket **32** can be comprised of a water- and rust-resistant material such as stainless steel.

[0112] In some embodiments, the hanger bracket **32** can be secured to hinge plate **36** of the suction cup base assembly **30** using a hinge assembly **35**. The hinge assembly **35** permits the hanger bracket **32** to extend as shown in FIG. **9**, and retract, as shown in FIG. **4**, thereby facilitating the compact storage or transport of the mirror assembly **10**, in some embodiments. The hinge assembly

35 can include a roll pin to support the hanger bracket **32** when it is used to hang the mirror assembly **10** from a door, ledge, or the like, in some embodiments. In some embodiments, the hinge assembly **35** can use a solid pin or set screws to support the hanger bracket **32** when it is used to hang the mirror assembly **10** from a door, ledge, or the like. A physical stop can be molded into hinge plate **36** to prevent the hanger bracket **32** from rotating further than 90° relative to the suction cup base assembly **30**, in some embodiments.

[0113] In some embodiments, a telescoping rod **40** extends between the suction cup base assembly **30** and the mirror **20**. In some embodiments, the first end **41** of the telescoping rod **40** is secured to the housing **27** of the mirror **20** using a ball joint. The ball joint allows the user of the mirror assembly **10** to pivot and adjust the mirror **20** to obtain the desired position, in some embodiments. In some embodiments, the second end **42** of the telescoping rod **40** is secured to the housing portion **31** of the suction cup base assembly **30**, by, for example, a swivel joint such that the suction cup base assembly **30** can be stored against the telescoping rod **40** when the mirror assembly **10** is not in use, thereby facilitating the compact storage or transport of the mirror assembly **10**. The telescoping rod **40** can be retracted or extended in order to adjust the height of the mirror **20** or distance from the user or to facilitate the compact storage or transport of the mirror assembly **10**, in some embodiments.

[0114] In some embodiments, a smooth disc of any suitable size (e.g., 5″ in diameter) can be provided to enhance the stability of the mirror assembly **10** when the telescoping rod **40** is fully extended by removably mounting the suction cup base assembly **30** to the disc using the suction cup **34**, thereby providing a larger footprint and increased stability to the mirror assembly **10**.

[0115] FIGS. **19-39** illustrate an example mirror assembly **110** in accordance with some embodiments. As shown in FIGS. **19** and **22**, in some embodiments, mirror assembly **110** comprises a mirror **120**, a suction cup base assembly **130**, and a multi-stage telescoping rod **140**. Mirror **120** can have a first reflective surface **121** on a first side and a second reflective surface **122** on a second side, in some embodiments. In some embodiments, there can be one or more light sources **123** disposed at a periphery of first reflective surface **121** or second reflective surface **122**. The light source or sources **123** can be powered by a battery (e.g., a rechargeable battery or batteries) or can be plugged into an electrical outlet, for example, in some embodiments. Rechargeable batteries can be located in suction cup base assembly **130**, in some embodiments. A battery retainer plate can hold rechargeable batteries in a pyramid configuration to provide room for a sliding tray that holds hanger bracket **132**, in some embodiments.

[0116] Mirror assembly **110** can include a switch **124** for activating or deactivating light source or sources **123**, for increasing the intensity of the light emitted by the light source, and for decreasing the intensity of the light emitted by light source **123**, in some embodiments. As shown in FIG. **19**, in some embodiments, switch **124** can be located in suction cup base assembly **130**. Light source or sources **123** can include a diffuser screen, in some embodiments. Housing **127** can be secured to a fourth stage **147** of multi-stage telescoping rod **140** using swivel joint **125**, in some embodiments. In some embodiments, mirror **120** can also have a lever **126** that when triggered rotates reflective surfaces **121** and **122** relative to light source or sources **123** about an axis normal to the vertical axis of multi-stage telescoping rod **140**, for example, thereby allowing a user to view the reflective surface that provides the desired magnification level. Mirror **120** can include a clamp brake that adds friction to the rotation of the first and second reflective surfaces, in some embodiments. In some embodiments, any other suitable one or more methods and axes of rotation can be used to rotate reflective surfaces **121** and **122**. In some embodiments, each reflective surface **121** and **122** can be non-magnifying or can be magnifying, and each can provide level a of magnification of, for example, 1×, 3×, 5×, 7×, 10×, 15×, or 20×.

[0117] Suction cup base assembly **130** can be substantially hemispherical in shape, comprising a round, flexible, concave diaphragm **134** (i.e., a suction cup) on the front side of the planar surface of the hemisphere, in some embodiments. In some embodiments, the suction cup base assembly

130 can comprise any suitable number of suction cups. In some embodiments, the suction cup base assembly **130** can comprise only one suction cup **134** in order to reduce the size and weight of the mirror assembly **110**, thereby increasing portability. The suction cup **134** can be used to removably mount the mirror assembly **110** to a horizontal or vertical surface, in some embodiments. The suction cup **134** can be a locking suction cup or a non-locking suction cup, in some embodiments. In some embodiments, a housing portion **131** is disposed on the reverse side of the planar surface of the hemisphere. The housing portion **131** can be made of any suitable material(s), such as a water-, abrasion-, and stain-resistant material, such as plastic, in some embodiments. In some embodiments, it may be desirable to use a material such as plastic for housing portion **131** since it will resist discoloration and rust that can occur due to conditions in which the mirror assembly can be utilized, e.g., in a damp environment such as a bathroom, adjacent to a water source such as a sink, or near substances that can stain or discolor such as makeup or other grooming products. In some embodiments, the plastic material from which the housing portion **131** of the suction cup base assembly **130** can be comprised can also be flexible such that it will not crack or break if the mirror assembly **110** is dropped. If the suction cup **134** is a locking suction cup, the housing portion **131** of the suction cup base assembly **130** can also include a trigger arm **133** for engaging the suction cup **134** such that it becomes removably mounted to a horizontal or vertical surface, in some embodiments. In some embodiments, trigger arm **133** can be comprised of any suitable material. In some embodiments, due to the conditions in which the mirror assembly **110** can be utilized, e.g., in a damp environment such as a bathroom, or adjacent to a water source such as a sink, the trigger arm **133** can be chosen to be comprised of a water- and rust-resistant material such as stainless steel or aluminum.

[0118] The suction cup base assembly **130** can also include a hanger bracket **132**, in some embodiments. The hanger bracket **132** provides support for the mirror assembly **110** to be hung in an inverted manner on a door, a ledge, or the like, in some embodiments. To permit the mirror assembly **110** to be hung in an inverted manner from a door, ledge, or the like, the hanger bracket **132** can have an internal dimension sufficient to accommodate the minimum and maximum standard thickness of an interior door, in some embodiments. In some embodiments, hanger bracket **132** can be shaped as shown, for example, in FIGS. **27**, **28**, and **30**A, or can be shaped in any manner suitable to allow it to support the mirror assembly **110** over a door, ledge, or the like. Hanger bracket **132** can be made of any suitable material, in some embodiments. For example, in some embodiments, the hanger bracket can be made of metal, such as stainless steel, so that it can adequately support the mirror assembly **110** and is preferably thin enough to permit the door on which the mirror assembly **110** can be hung to be closed. In some embodiments due to the conditions in which the mirror assembly **110** can be utilized, e.g., in a damp environment such as a bathroom, or adjacent to a water source such as a sink, hanger bracket **132** can be comprised of a water- and rust-resistant material such as stainless steel.

[0119] In some embodiments, hanger bracket **132** can be held in place by a sliding tray, thereby allowing hanger bracket **132** to slide in and out of suction cup base assembly **130**. In some embodiments, the sliding tray permits hanger bracket **132** to extend, as show in FIG. **27**, and retract, as shown in FIG. **26**, thereby facilitating the compact storage or transport of mirror assembly **110**.

[0120] In some embodiments, a multi-stage telescoping rod **140** can extend between suction cup base assembly **130** and mirror **120**. In some embodiments, a first end **141** of multi-stage telescoping rod **140** can be secured to housing **127** of mirror **120** using swivel joint **125**. Swivel joint **125** can allow the user of mirror assembly **110** to adjust mirror **120** to obtain the desired position, in some embodiments. A second end **142** of multi-stage telescoping rod **140** can be attached to housing portion **131** of suction cup base assembly **130**, in some embodiments. In some embodiments, telescoping rod **140** can be retracted or extended in order to adjust the height of mirror **120** or distance from the user or to facilitate the compact storage or transport of mirror assembly **110**. Multi-stage telescoping rod **140** can have a base stage **143** and four additional stages, wherein base stage **143** has a certain diameter, stage one **144** and stage two **145** have progressively decreasing diameters, and stage three **146** and stage four **147** have progressively increasing diameters such that the diameter of stage four **147** can be equal to the diameter of base stage **143**, in some embodiments.

[0121] In some embodiments, a smooth disc of any suitable size (e.g., 5″ in diameter) can be provided to enhance the stability of the mirror assembly **110** when the telescoping rod **140** is fully extended by removably mounting the suction cup base assembly **130** to the disc using the suction cup **134**, thereby providing a larger footprint and increased stability to the mirror assembly **110**.

[0122] In some embodiments, a mirror assembly comprises a mirror, a suction cup base assembly, and a telescoping rod. The mirror can be substantially hemispherical in shape, comprising a flat, circular reflective surface disposed on the front side of the planar surface of the hemisphere, and a housing disposed on the reverse side of the planar surface of the hemisphere, in some embodiments. The housing can have a socket for use in a ball joint by which a first end of a telescoping rod can be secured to the housing or the housing can be secured to the first end of a telescoping rod using a swivel joint, in some embodiments. In some embodiments, there can be one or more light sources disposed at the periphery of the reflective surface. The light source or sources can be powered by a battery (e.g., a rechargeable battery) or can be plugged into an electrical outlet, for example, in some embodiments. The light source or sources can be activated and deactivated by a switch, or the like, which can be located adjacent the light source on the front side of the planar surface of the hemisphere or on the reverse side, or can be located in the suction cup base, for example, in some embodiments. In some embodiments, the reflective surface can be non-magnifying or can be magnifying, providing magnification at a level of, for example, 1×, 3×, 5×, 7×, 10×, 15×, or 20×.

[0123] FIGS. **40-68** illustrate further details of a telescoping rod that can be used in place of the telescoping rods described above in accordance with some embodiments.

[0124] Turning to FIG. **40**, a core telescope assembly **400** of the telescoping rod is shown in accordance with some embodiments. As illustrated, assembly **400** can include an upper arm **402**, two pulleys **404**, a belt **406**, two clamps **408**, and a lower arm **410**, in some embodiments. As shown, belt **406** wraps around the two pulleys and is clamped to each of arms **402** and **410** by a respective clamp **408**, in some embodiments.

[0125] Arms **402** is further illustrated in FIG. **41**, in accordance with some embodiments. The arm can have any suitable dimensions (e.g., such as a length of 10 9/16″, a height of 0.425″, and a thickness of 0.145″) and can have any suitable cross-section, such as the cross-section shown in the figure, and other features, in some embodiments. The arm can be made of any suitable material, such as injection-molded plastic, in some embodiments. In some embodiments, the arm can be made of glass-filled, injection-molded plastic.

[0126] FIG. **42** illustrates an example of belt **406**, in accordance with some embodiments. As shown, belt **406** can be a toothed belt, in some embodiments. In some embodiments, belt **406** can be toothless (smooth) and have any suitable cross-section. In some embodiments, belt **406** can have any suitable dimensions, such as: a width of ⅛″, 4 mm, or any other suitable value; a pitch is 0.080″ (approx. 2 mm) or any other suitable value; and a length suitable to wrap around the two pulleys. In some embodiments, when implemented with teeth, belt **406** can have any suitable number of teeth, such as **235** teeth or any other suitable number. In some embodiments, belt **406** can be made of any suitable materials, such as urethane. In some embodiments, belt **406** can be replaced with another object that performs a similar function, such as a chain (e.g., similar in structure to a bicycle chain), a cable, a wire, a rope, etc.

[0127] FIG. **43** illustrates an example of clamp **408** in accordance with some embodiments. As shown, clamp **408** can include a suitable carved-out section to receive a portion of belt **406**, in some embodiments. When belt **406** is implemented with teeth, clamp **408** can include suitable complementary teeth to engage the teeth on the belt, in some embodiments. When implemented with a toothless belt or the belt is replaced with a cable, wire, or rope, clamp **408** can include any suitable surface texture (such as a ridged or jagged texture) to cause the belt, cable, wire, or rope not to slip with respect to the clamp, in some embodiments. As also shown, in some embodiments, the clamp can include holes so that the clamp can be attached to an arm via any suitable fastener, such as a screw, rivet, or other fastener. In some embodiments, the clamp can be glued or welded to the arm. In some embodiments, the clamp can have any suitable dimensions. The clamp can be made of any suitable material, such as injection-molded plastic, in some embodiments. In some embodiments, the clamp can be made of glass-filled, injection-molded plastic.

[0128] FIG. **44** illustrates an example of pulley **404** in accordance with some embodiments. As shown, pulley **404** can include a suitable carved-out section to receive a portion of belt **406**, in some embodiments. When belt **406** is implemented with teeth, pulley **404** can include suitable complementary teeth to engage the teeth on the belt, in some embodiments. When implemented with a toothless belt or the belt is replaced with a cable, wire, or rope, pulley **404** can include any suitable surface texture in the portion that engages the belt, cable, wire, or rope, in some embodiments. As also shown, in some embodiments, the pulley can include a hole so that the pulley can be fixed in position in the telescoping rod using any suitable fastener, such as a screw or pin (e.g., spring pin or non-spring pin). In some embodiments, the pulley can have any suitable dimensions. The pulley can be made of any suitable material, such as injection-molded plastic, in some embodiments. In some embodiments, the pulley can be made of glass-filled, injection-molded plastic. In some embodiments, as shown in FIG. **70**, pulley **404** can include a groove for an o-ring **405** that can press against a corresponding rod **402**/**410** to which the pulley is adjacent. This can cause friction between the pulley and the rod and slow the rate at which the pulley rotates, in some embodiments. In some embodiments, to further slow the rate at which the pulley rotates, bumps **407**, and/or indentations, can be located on/in rod **402**/**410** so that the o-ring rolls over the bumps and/or in and out of the indentations. Such bumps and/or indentations can have any suitable size and spacing, in some embodiments.

[0129] Arms **410** is further illustrated in FIG. **45**, in accordance with some embodiments. The arm can have any suitable dimensions (e.g., such as a length of 10 9/16″, a height of 0.425″, and a thickness of 0.145″) and can have any suitable cross-section, such as the cross-section shown in the figure, and any other features, in some embodiments. The arm can be made of any suitable material, such as injection-molded plastic, in some embodiments. In some embodiments, the arm can be made of glass-filled, injection-molded plastic.

[0130] Turning to FIG. **46**, an illustration of assembly **400** (including arms **402** and **410**) inside a core tube **412** is shown, in accordance with some embodiments. More detailed illustrations of core tube **412** are shown in FIG. **47**, in accordance with some embodiments. As illustrated, core tube **412** can include holes **413** through which pins or screws can be placed to secure pulleys with respect to the core tube. The core tube can have any suitable dimensions and can have any suitable cross-section, such as the cross-section shown in the figure, and any other features, in some embodiments. The core tube can be made of any suitable material, such as extruded aluminum with secondary machined features, in some embodiments.

[0131] FIG. **48** shows the components of FIG. **46** with the addition of an end cap **414**, in accordance with some embodiments. End cap **414** is further illustrated in FIG. **49** in some embodiments. End cap **414** can have any suitable shape and dimensions, such as the shape shown in the figure and can be made of any suitable material, such as injection molded Delrin, in some embodiments. In some embodiments, end cap **414** can be sized to have an outer surface that slides on an inside surface of a bottom tube described below.

[0132] FIG. **50** shows the components of FIG. **48** with the addition of a cable **416**, in accordance with some embodiments. As shown cable **416** can sit on top of core tube **412** and pass through a hole in end cap **414**. Cable **416** can be formed from any suitable material and have any suitable dimensions, in some embodiments. As shown, cable **416** can be coiled to allow the cable to stretch between the minimum length of the telescoping rod and the maximum length of the telescoping rod. In some embodiments, rather than being coiled, cable **406** can be non-coiled and formed from a material that can be suitably stretched.

[0133] FIG. **51** shows the components of FIG. **50** inside a middle tube **418**, in accordance with some embodiments. More detailed illustrations of middle tube **418** are shown in FIG. **52**, in accordance with some embodiments. The middle tube can have any suitable dimensions (e.g., a length of 10 13/16″ and an outside diameter of 1.10″) and can have any suitable cross-section, such as the cross-section shown in the figure, and any other features, in some embodiments. The middle tube can be made of any suitable material, such as extruded aluminum with secondary machined features, in some embodiments.

[0134] FIG. **53** shows the components of FIG. **51** with a top tube **420**. More detailed illustrations of top tube **420** are shown in FIG. **54**, in accordance with some embodiments. The top tube can have any suitable dimensions (e.g., a length of 10 1/16″ and an outside diameter of 0.930″) and can have any suitable cross-section, such as the cross-section shown in the figure, and any other features, in some embodiments. The top tube can be made of any suitable material, such as extruded aluminum with secondary machined features, in some embodiments.

[0135] FIG. **55** shows an illustration of top tube **420** of FIG. **54** along with a bushing **422**, in accordance with some embodiments. Bushing **422** is further illustrated in FIG. **56** in some embodiments. Bushing **422** can have any suitable shape and dimensions, such as the shape shown in the figure and can be made of any suitable material, such as injection molded Delrin, in some embodiments. In some embodiments, as shown, bushing **422** can be formed from two half rings, and each half ring can include rivets **423** that can be flattened when end cap half rings are placed on top tube **420** so that the half rings are secure to the top tube. In some embodiments, as shown in FIG. **72**, a bushing **425** can be used in place of bushing **422**. As shown, bushing **425** has a diameter in a middle portion that is wider than other portions of the bushing. This can cause the bushing to apply friction to the inner surface of the middle tube and thereby slow movement between the top tube and the middle tube.

[0136] FIG. **57** shows an illustration of the components of FIG. **55** along with a bushing **424**, in accordance with some embodiments. Bushing **424** is further illustrated in FIG. **58** in accordance with some embodiments. Bushing **424** can have any suitable shape and dimensions, such as the shape shown in the figure and can be made of any suitable material, such as injection molded Delrin, in some embodiments. In some embodiments, instead of bushing **424**, bushing **426** shown in FIG. **60** (and described in connection therewith) can be used in the same location as shown for bushing **424**. When bushing **426** is used, as shown in FIG. **71**, a raised portion **427** of the bushing can press on an inside surface of middle tube **418**. This causes a finger of bushing **426** to press against an outside surface of top tube **420**, thereby slowing movement between the middle tube and the top tube.

[0137] FIG. **59** shows an illustration of the components of FIG. **57** along with a bushing **426**, in accordance with some embodiments. Bushing **426** is further illustrated in FIG. **60** in accordance with some embodiments. Bushing **426** can have any suitable shape and dimensions, such as the shape shown in the figure and can be made of any suitable material, such as injection molded Delrin, in some embodiments. When bushing **426** is used, as shown in FIG. **71**, a raised portion **427** of the bushing can press on an inside surface of a bottom tube **430** (described below). This causes a finger of bushing **426** to press against an outside surface of middle tube **418**, thereby slowing movement between the middle tube and the bottom tube. In some embodiments, instead of bushing **426**, bushing **424** shown in FIG. **58** (and described in connection therewith) can be used in the same location as shown for bushing **426**.

[0138] FIG. **61** shows an illustration of a friction pad **428** shown in end cap **414**, in accordance with some embodiments. Friction pad **428** is further illustrated in FIG. **62** in accordance with some

embodiments. Friction pad **428** can have any suitable shape and dimensions, such as the shape shown in the figure and can be made of any suitable material, such as a polymer (e.g., nylon, aliphatic polyamide, thermoplastic polyurethane, silicone), metal (e.g., brass, copper), a composite (e.g., fiberglass, carbon fiber), a ceramic, and/or any other suitable material that can provide a consistent friction value, in some embodiments. Friction pad **428** can be configured to rub against bottom tube **430** shown in FIGS. **63** and **64** and described below, and slow the rate at which the telescoping rod lengthens and shortens. In some embodiments, the pad is longer in a first dimension along the axis of the middle tube than it is wide in a second dimension that is perpendicular to the first dimension.
[0139] FIG. **63** shows an illustration of top tube **420** and middle tube **418** along with bottom tube **430**, in accordance with some embodiments. More detailed illustrations of bottom tube **430** are shown in FIG. **64**, in accordance with some embodiments. The bottom tube can have any suitable dimensions (e.g., a length of 11 11/16″ and an outside diameter of 1.295″) and can have any suitable cross-section, such as the cross-section shown in the figure, and any other features, in some embodiments. The top tube can be made of any suitable material, such as extruded aluminum with secondary machined features, in some embodiments.
[0140] FIG. **65** shows the components of FIG. **63** with the addition of an end cap **432**, in accordance with some embodiments. End cap **432** is further illustrated in FIG. **66** in some embodiments. End cap **432** can have any suitable shape, such as the shape shown in the figure, dimensions, and can be made of any suitable material, such as injection molded Delrin, in some embodiments. Clips on the sides of end cap **432** can attach the telescoping rod to a ball joint or swivel joint at the back of mirror **20** or **120** described above.
[0141] FIG. **67** shows the components of FIG. **65** with the addition of an end cap **434**, in accordance with some embodiments. End cap **434** is further illustrated in FIG. **68** in some embodiments. End cap **434** can have any suitable shape, such as the shape shown in the figure, dimensions, and can be made of any suitable material, such as injection molded Delrin, in some embodiments. Holes in end cap **434** can connect the telescoping rod to the base assembly **30** or **130**, described above.
[0142] FIG. **69** shows an example of a suction cup **34**/**134** that can be used as suction cup **34** or **134** described above. As shown, suction cup **34**/**134** can include a bump **234**. When the center of the suction cup is pulled away from the surface on which it is located, a vacuum is created in the suction cup and it is held to the surface. In this position, the bump is also pulled away from the surface. However, when the center of the suction cup is released and allowed to move toward the surface, the bump presses against the surface and causes the seal formed by the perimeter of the suction cup to the surface to be broken and the vacuum of the suction cup released.
[0143] It will be understood by persons skilled in the art that modifications can be made to the embodiments described herein while remaining within the scope of the claimed invention.

Claims

1. A telescoping rod, comprising: a core tube having a first end and a second end; a first pulley mounted inside the core tube at the first end; a second pulley mounted inside the core tube at the second end; a belt mounted around and between the first pulley and the second pulley; a first arm coupled to the belt, partially located inside the core tube, and extending outward of the first end of the core tube; and a second arm coupled to the belt, partially located inside the core tube, and extending outward of the second end of the core tube.
2. The telescoping rod of claim 1, wherein the belt contains teeth in an inside surface of the belt, the first pulley contains teeth that engage the teeth of the belt, and the second pulley contains teeth that engage the belt.
3. The telescoping rod of claim 1, further comprising a first clamp that couples the belt to the first

arm and a second clamp that couples the belt to the second arm.
4. The telescoping rod of claim 1, further comprising a middle tube that contains the core tube, wherein the middle tube has a first end and a second end.
5. The telescoping rod of claim 4, further comprising a cable positioned between to core tube and the middle tube.
6. The telescoping rod of claim 4, further comprising a top tube that at least partially fits inside the middle tube, extends outward from the middle tube at the first end of the middle tube, and is coupled to the first arm.
7. The telescoping rod of claim 6, further comprising first bushing located on an outside surface of the top tube at the second end of the top tube.
8. The telescoping rod of claim 7, wherein the first bushing comprises two parts.
9. The telescoping rod of claim 7, wherein a middle portion of the first bushing has a larger diameter than another portion of the first bushing.
10. The telescoping rod of claim 6, further comprising a second bushing located at the first end of the middle tube between the top tube and the middle tube.
11. The telescoping rod of claim 10, wherein the second bushing includes at least one raised surface that presses on an inside surface of the middle tube and cause a portion of the second bushing to bend and apply pressure to an outside surface of the top tube.
12. The telescoping rod of claim 6, further comprising a top cap that couples the first arm and the top tube to a mirror.
13. The telescoping rod of claim 6, further comprising a bottom tube that at least partially extends over the middle tube, extends away from the middle tube at the second end of the middle tube, and is coupled to the second arm.
14. The telescoping rod of claim 13, further comprising a third bushing located at a first end of the bottom tube between the bottom tube and the middle tube.
15. The telescoping rod of claim 14, wherein the third bushing includes at least one raised surface that presses on an inside surface of the bottom tube and cause a portion of the third bushing to bend and apply pressure to an outside surface of the middle tube.
16. The telescoping rod of claim 13, further comprising a pad coupled to the middle tube that presses on the bottom tube to slow movement between the middle tube and the bottom tube.
17. The telescoping rod of claim 16, wherein the pad is longer in a first dimension along the axis of the middle tube than it is wide in a second dimension that is perpendicular to the first dimension.
18. The telescoping rod of claim 13, further comprising a bottom cap that couples the second arm and the bottom tube to a base.
